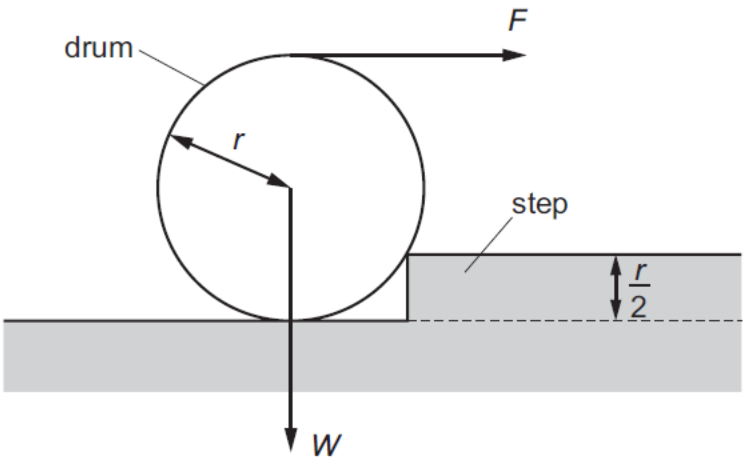
**6** A cylindrical drum has radius $r$ and weight $W$. The drum is to be rolled over onto a step of height $\frac{r}{2}$ by a horizontal force $F$ applied to the top of the drum.




What is the minimum force $F$ required for the drum to start rolling on the step?

**A** $\frac{W}{2}$ **B** $\frac{W}{\sqrt{3}}$ **C** $W$ **D** $\frac{\sqrt{3}}{2}W$

Ans: **B**

Taking moments about the step,

$$W\left(\frac{\sqrt{3}}{2}r\right)=F\left(2r-\frac{r}{2}\right)$$

$$F=\frac{W\left(\frac{\sqrt{3}}{2}r\right)}{\frac{3r}{2}}$$

$$=\frac{W}{\sqrt{3}}$$

**7** A ball of mass m falls freely from rest as shown below. When it has reached a speed v, it just